

# COMPSCI 1DM3 - Winter 2026

## Discrete Mathematics for Computer Science

Angela Zavaleta Bernuy



# Midterm

- Date: Monday, February 23, 2025
- Time: **7:00 pm - 9:00 pm**
- Location: PGCLL B138

# Last Lecture

- Given  $p \rightarrow q$ , we defined
  - **Converse:**  $q \rightarrow p$
  - **Contrapositive:**  $\neg q \rightarrow \neg p$ .
  - **Inverse:**  $\neg p \rightarrow \neg q$
- $p$  and  $q$  are equivalent ( $p \equiv q$ ) if they have the same truth values
- How to show equivalence of propositions? By using logical equivalence laws, such as
  - distributive laws:  $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
  - De Morgan's laws:  $\neg(p \vee q) \equiv \neg p \wedge \neg q$
  - Absorption laws:  $p \vee (p \wedge q) \equiv p$
  - Implication law:  $p \rightarrow q \equiv \neg p \vee q$
  - $p \rightarrow q \equiv \neg q \rightarrow \neg p$
- Tautology is a proposition that is always true!
- $p \equiv q$  if and only if  $p \leftrightarrow q$  is a tautology!

# Last Lecture: Tautology

**Example.** Determine whether  $(\neg p \wedge (p \rightarrow q)) \rightarrow p$  is a tautology.

No. We need to pick a particular truth values for  $p$  and  $q$  making the proposition false. To see this, recall that the implication is false when its hypothesis is true and its conclusion is false. So, we want  $p$  to be false.

Now, let's look at the hypothesis, i.e.,  $(\neg p \wedge (p \rightarrow q))$ . Since  $p$  is false, both  $\neg p$  and  $p \rightarrow q$  are true, thus the hypothesis is true.

Thus, if  $p$  is false, then the hypothesis is true while the conclusion is false, implying that the implication  $(\neg p \wedge (p \rightarrow q)) \rightarrow p$  is false. So it can't be a tautology.

# Last Lecture: Tautology

**Example.** Determine whether  $(\neg p \wedge (p \rightarrow q)) \rightarrow \neg q$  is a tautology.

① simplify  $\rightsquigarrow$  T ✓

② Check when this F

# Last Lecture: Tautology

**Example.** Determine whether  $(\neg p \wedge (p \rightarrow q)) \rightarrow \neg q$  is a tautology.

No. Suppose  $p$  is false and  $q$  is true.

$$\neg p \wedge (p \rightarrow T) \equiv T$$

$$\neg q \equiv F$$
$$q \equiv T$$

$$\neg p \equiv T, \quad p \rightarrow T \equiv T$$

$$p \equiv F$$

# Propositional function

- Recall that  $x > 3$  is not a proposition. Because, its truth value depends on  $x$ .

# Propositional function

- Recall that  $x > 3$  is not a proposition. Because, its truth value depends on  $x$ .
  - ▶ A **proposition** is a declarative sentence with a true or false value (**but not both!**).
- Can we define a *proposition* that depends on a variable  $x$ ?

# Propositional function

- Recall that  $x > 3$  is not a proposition. Because, its truth value depends on  $x$ .
  - ▶ A **proposition** is a declarative sentence with a true or false value (**but not both!**).
- Can we define a *proposition* that depends on a variable  $x$ ?
- Let  $P(x)$  be the statement “ $x$  is greater than 3”. Then  $P(x)$  is a proposition for a *given*  $x$ ; e.g.,  $P(4)$  is T and  $P(2)$  is F.

# Propositional function

- Recall that  $x > 3$  is not a proposition. Because, its truth value depends on  $x$ .
  - ▶ A **proposition** is a declarative sentence with a true or false value (but not both!).
- Can we define a *proposition* that depends on a variable  $x$ ?
- Let  $P(x)$  be the statement “ $x$  is greater than 3”. Then  $P(x)$  is a proposition for a given  $x$ ; e.g.,  $P(4)$  is T and  $P(2)$  is F.
- Propositional functions may even depend on several variables: Let  $Q(x, y)$  be “ $x^2 + y^2 = 1$ ”. Note that  $Q(x, y)$  is a proposition for any given  $x$  and  $y$ :  $Q(1, 0)$  is T and  $Q(10, 10)$  is F.

# Propositional function

- Recall that  $x > 3$  is not a proposition. Because, its truth value depends on  $x$ .
  - ▶ A **proposition** is a declarative sentence with a true or false value (but not both!).
- Can we define a *proposition* that depends on a variable  $x$ ?
- Let  $P(x)$  be the statement “ $x$  is greater than 3”. Then  $P(x)$  is a proposition for a given  $x$ ; e.g.,  $P(4)$  is T and  $P(2)$  is F.
- Propositional functions may even depend on several variables: Let  $Q(x, y)$  be “ $x^2 + y^2 = 1$ ”. Note that  $Q(x, y)$  is a proposition for any given  $x$  and  $y$ :  $Q(1, 0)$  is T and  $Q(10, 10)$  is F.
- With this definition, we can represent mathematical statements.

# Quantifiers

- We know that  $x^2$  is always non-negative. Let  $P(x)$  be " $x^2 \geq 0$ ". Then,  $P(x)$  is T for *all* real numbers  $x$ .

# Quantifiers

- We know that  $x^2$  is always non-negative. Let  $P(x)$  be " $x^2 \geq 0$ ". Then,  $P(x)$  is T for *all* real numbers  $x$ .
- This statement has the quantifier "for all".

# Quantifiers

- We know that  $x^2$  is always non-negative. Let  $P(x)$  be " $x^2 \geq 0$ ". Then,  $P(x)$  is T for *all* real numbers  $x$ .
- This statement has the quantifier "for all".

## Definition

The *universal quantification* of  $P(x)$  is the proposition:

$P(x)$  for all values of  $x$  in the domain

and is denoted by  $\forall x P(x)$ .

# Quantifiers

- We know that  $x^2$  is always non-negative. Let  $P(x)$  be " $x^2 \geq 0$ ". Then,  $P(x)$  is T for *all* real numbers  $x$ .
- This statement has the quantifier "for all".

## Definition

The *universal quantification* of  $P(x)$  is the proposition:

$P(x)$  for all values of  $x$  in the domain

and is denoted by  $\forall x P(x)$ .

**Example.** Let  $P(x)$  be  $x^2 \geq 0$  and the domain is real numbers. What is the truth value of  $\forall x P(x)$ ?

T

# Universal quantifier

To show  $\forall x P(x)$  is true, we need to show that  $P(x)$  is true for all  $x$  *in the domain*.

# Universal quantifier

To show  $\forall x P(x)$  is true, we need to show that  $P(x)$  is true for all  $x$  in the domain.

**Example.** Is  $\forall x(2x \geq x)$  true?

$$\mathbb{R} \quad x = -2 \quad -4 \geq -2 \quad (\text{F})$$

$$\mathbb{N} \quad (\text{T})$$

# Universal quantifier

To show  $\forall x P(x)$  is true, we need to show that  $P(x)$  is true for all  $x$  *in the domain*.

Example. Is  $\forall x(2x \geq x)$  true?

Example. Is  $\forall x(x^2 \geq x)$  true?

$\mathbb{R}$

# Universal quantifier

To show  $\forall x P(x)$  is true, we need to show that  $P(x)$  is true for all  $x$  *in the domain*.

**Example.** Is  $\forall x(2x \geq x)$  true?

**Example.** Is  $\forall x(x^2 \geq x)$  true?

Sometimes we write  $\forall x > 1(x^2 \geq x)$  to compactly specify the domain!  $\mathbb{R}$

$\forall x$

# Existential quantifier

## Definition

The *existential quantification* of  $P(x)$  is the proposition:

**There exists an  $x$  in the domain such that  $P(x)$**   
and is denoted by  $\exists x P(x)$ .

# Existential quantifier

## Definition

The *existential quantification* of  $P(x)$  is the proposition:

**There exists an  $x$  in the domain such that  $P(x)$**   
and is denoted by  $\exists x P(x)$ .

To show  $\exists x P(x)$  is true, we need to show that there exists *at least one  $x$  in the domain* such  $P(x)$  is true.

# Existential quantifier

## Definition

The *existential quantification* of  $P(x)$  is the proposition:

There exists an  $x$  in the domain such that  $P(x)$   
and is denoted by  $\exists x P(x)$ .

To show  $\exists x P(x)$  is true, we need to show that there exists *at least one*  $x$  in the domain such  $P(x)$  is true.

**Example.** Is  $\exists x(2x \geq x)$  true if the domain is negative reals?

~~$x = -2$~~

$-4 \geq -2$

$-2 \geq -1$

$x = -\frac{1}{2}$

$\downarrow$   
 ~~$-1 \geq -\frac{1}{2}$~~

$\nearrow$  No

**Example.** Is  $\exists x(x^2 \geq x)$  true if the domain is reals?

$x = 0$

$\top$

# Negation of quantifiers

Question.

What is the negation of  $\forall x P(x)$ ?

# Negation of quantifiers

## Question.

What is the negation of  $\forall x P(x)$ ?

Answer: (regardless of the domain)

$$\neg(\forall x P(x)) \equiv \exists x \neg P(x).$$

 **Proof.** Suppose  $\neg \forall x P(x)$  is F. Then,  $P(x)$  is T for all  $x$  in the domain. Thus, there can't exist an  $x$  such that  $\neg P(x)$  is T, implying  $\exists x \neg P(x)$  is F.

# Negation of quantifiers

## Question.

What is the negation of  $\forall x P(x)$ ?

Answer: (regardless of the domain)

$$\neg \forall x P(x) \equiv \exists x \neg P(x).$$

**Proof.** Suppose  $\neg \forall x P(x)$  is F. Then,  $P(x)$  is T for all  $x$  in the domain. Thus, there can't exist an  $x$  such that  $\neg P(x)$  is T, implying  $\exists x \neg P(x)$  is F.

Now we need to show that if  $\neg \forall x P(x)$  is T then so is  $\exists x \neg P(x)$ . Let's show its contrapositive (Recall:  $p \rightarrow q \equiv \neg q \rightarrow \neg p$ ):

# Negation of quantifiers

## Question.

What is the negation of  $\forall x P(x)$ ?

Answer: (regardless of the domain)

$$\neg \forall x P(x) \equiv \exists x \neg P(x).$$

**Proof.** Suppose  $\neg \forall x P(x)$  is F. Then,  $P(x)$  is T for all  $x$  in the domain. Thus, there can't exist an  $x$  such that  $\neg P(x)$  is T, implying  $\exists x \neg P(x)$  is F.

Now we need to show that if  $\neg \forall x P(x)$  is T then so is  $\exists x \neg P(x)$ . Let's show its contrapositive (Recall:  $p \rightarrow q \equiv \neg q \rightarrow \neg p$ ):

Suppose  $\exists x \neg P(x)$  is F, meaning, there doesn't exist any  $x$  such that  $\neg P(x)$  is T.

Thus,  $P(x)$  is T for all  $x$  in the domain, meaning  $\forall x P(x)$  is T and hence,  $\neg \forall x P(x)$  is F.

# Negation of quantifiers

## Question.

What is the negation of  $\forall x P(x)$ ?

Answer: (regardless of the domain)

$$\neg \forall x P(x) \equiv \exists x \neg P(x).$$

**Proof.** Suppose  $\neg \forall x P(x)$  is F. Then,  $P(x)$  is T for all  $x$  in the domain. Thus, there can't exist an  $x$  such that  $\neg P(x)$  is T, implying  $\exists x \neg P(x)$  is F.

Now we need to show that if  $\neg \forall x P(x)$  is T then so is  $\exists x \neg P(x)$ . Let's show its contrapositive (**Recall:  $p \rightarrow q \equiv \neg q \rightarrow \neg p$** ):

Suppose  $\exists x \neg P(x)$  is F, meaning, there doesn't exist any  $x$  such that  $\neg P(x)$  is T. Thus,  $P(x)$  is T for all  $x$  in the domain, meaning  $\forall x P(x)$  is T and hence,  $\neg \forall x P(x)$  is F.

**Example.**  $\neg \forall x (x^2 > x) \equiv \exists x (x^2 \leq x)$ .

# Negation of quantifiers

Question.

What is the negation of  $\exists x P(x)$ ?

# Negation of quantifiers

## Question.

What is the negation of  $\exists x P(x)$ ?

Answer: (regardless of the domain)

$$\neg \exists x P(x) \equiv \forall x \neg P(x).$$

**Proof.** Based on the previous result,  $\neg \forall x \neg P(x) = \exists x P(x)$ . Therefore,  
 $\neg \exists x P(x) \equiv \neg(\exists x P(x)) \equiv \neg(\neg \forall x \neg P(x)) \equiv \forall x \neg P(x).$

# Example

Show that  $\neg \forall x (P(x) \rightarrow Q(x)) \equiv \exists x (P(x) \wedge \neg Q(x))$ .

$$\begin{aligned} \text{LHS} &\equiv \exists x \neg [P(x) \rightarrow Q(x)] \\ &\equiv \exists x \neg [\neg P(x) \vee Q(x)] && \text{imp. l.} \\ &\equiv \exists x [\neg \neg P(x) \wedge \neg Q(x)] && \text{D. M.} \\ &\equiv \exists x (P(x) \wedge \neg Q(x)) \end{aligned}$$

□

## Example

$$P \rightarrow Q \equiv \neg P \vee Q$$

Recall  $\neg \forall x (P(x) \rightarrow Q(x)) \equiv \exists x (P(x) \wedge \neg Q(x))$ .

# Example

Recall  $\neg \forall x (P(x) \rightarrow Q(x)) \equiv \exists x (P(x) \wedge \neg Q(x))$ .

Let the domain be people in a class.

# Example

Recall  $\neg\forall x(P(x) \rightarrow Q(x)) \equiv \exists x(P(x) \wedge \neg Q(x))$ .

Let the domain be people in a class.

- $P(x)$ : " $x$  is a student"

# Example

Recall  $\neg\forall x(P(x) \rightarrow Q(x)) \equiv \exists x(P(x) \wedge \neg Q(x))$ .

Let the domain be people in a class.

- $P(x)$ : " $x$  is a student"
- $Q(x)$ : " $x$  has an ID card"

$\forall x(P(x) \rightarrow Q(x))$ : "Every student has an ID card."

# Example

Recall  $\neg\forall x(P(x) \rightarrow Q(x)) \equiv \exists x(P(x) \wedge \neg Q(x))$ .

Let the domain be people in a class.

- $P(x)$ : " $x$  is a student"
- $Q(x)$ : " $x$  has an ID card"

$\forall x(P(x) \rightarrow Q(x))$ : "Every student has an ID card."

Negating it means:

$\neg\forall x(P(x) \rightarrow Q(x))$ : "Not every student has an ID card";

# Example

Recall  $\neg\forall x(P(x) \rightarrow Q(x)) \equiv \exists x(P(x) \wedge \neg Q(x))$ .

Let the domain be people in a class.

- $P(x)$ : " $x$  is a student"
- $Q(x)$ : " $x$  has an ID card"

$\forall x(P(x) \rightarrow Q(x))$ : "Every student has an ID card."

Negating it means:

$\neg\forall x(P(x) \rightarrow Q(x))$ : "Not every student has an ID card"; which is equivalent to:  
"There exists a person who is a student and does not have an ID card."

# Example

Recall  $\neg\forall x(P(x) \rightarrow Q(x)) \equiv \exists x(P(x) \wedge \neg Q(x))$ .

Let the domain be people in a class.

- $P(x)$ : " $x$  is a student"
- $Q(x)$ : " $x$  has an ID card"

$\forall x(P(x) \rightarrow Q(x))$ : "Every student has an ID card."

Negating it means:

$\neg\forall x(P(x) \rightarrow Q(x))$ : "Not every student has an ID card"; which is equivalent to:  
"There exists a person who is a student and does not have an ID card."

$(\exists x(P(x) \wedge \neg Q(x)))$

# Nested quantifiers

Let's use both quantifiers to obtain a proposition:

$$\forall x \exists y (x^2 - y^2 = 1).$$

This means “For all  $x$  in the domain, there exists a  $y$  in the domain such that  $x^2 - y^2 = 1$ .”

# Order matters

In propositions involving nested quantifiers, order of quantifiers might be important.

# Order matters

In propositions involving nested quantifiers, order of quantifiers might be important.

- $\forall x \forall y P(x, y)$  is the same as  $\forall y \forall x P(x, y)$

# Order matters

In propositions involving nested quantifiers, order of quantifiers might be important.

- $\forall x \forall y P(x, y)$  is the same as  $\forall y \forall x P(x, y)$
- $\exists x \exists y P(x, y)$  is the same as  $\exists y \exists x P(x, y)$

# Order matters

In propositions involving nested quantifiers, order of quantifiers might be important.

- $\forall x \forall y P(x, y)$  is the same as  $\forall y \forall x P(x, y)$
- $\exists x \exists y P(x, y)$  is the same as  $\exists y \exists x P(x, y)$
- However,  $\forall x \exists y P(x, y)$  is **NOT** the same as  $\exists y \forall x P(x, y)$ .

# Question

Let  $P(x, y)$  be any propositional function. Show that

$$\exists x \forall y P(x, y) \rightarrow \forall y \exists x P(x, y),$$

but NOT the other way around.

# Question

Let  $P(x, y)$  be any propositional function. Show that

$$\exists x \forall y P(x, y) \rightarrow \forall y \exists x P(x, y),$$

but NOT the other way around.

Suppose domain of  $x$  is all students enrolled in 1DM3 and domain of  $y$  is all lectures of 1DM3, and let

$P(x, y)$  = student  $x$  falls asleep during lecture  $y$ .

# Question

Let  $P(x, y)$  be any propositional function. Show that

$$\exists x \forall y P(x, y) \rightarrow \forall y \exists x P(x, y),$$

but NOT the other way around.

Suppose domain of  $x$  is all students enrolled in 1DM3 and domain of  $y$  is all lectures of 1DM3, and let

$$P(x, y) = \text{student } x \text{ falls asleep during lecture } y.$$

Note that  $\exists x \forall y P(x, y)$  means there exists a student who sleeps through all lectures (call them Snoozer). However,  $\forall y \exists x P(x, y)$  means in every lecture some student falls asleep. But there might be a different sleeper in each lecture (i.e., different from Snoozer).

# Question

Let  $P(x, y)$  be any propositional function. Show that

$$\exists x \forall y P(x, y) \rightarrow \forall y \exists x P(x, y),$$

but NOT the other way around.

# Question

Let  $P(x, y)$  be any propositional function. Show that

$$\exists x \forall y P(x, y) \rightarrow \forall y \exists x P(x, y),$$

but NOT the other way around.

Suppose  $\exists x \forall y P(x, y)$  is F, then the above implication is T (think of the truth table for  $p \rightarrow q$ ). This case is not interesting!

# Question

Let  $P(x, y)$  be any propositional function. Show that

$$\exists x \forall y P(x, y) \rightarrow \forall y \exists x P(x, y),$$

but NOT the other way around.

Suppose  $\exists x \forall y P(x, y)$  is F, then the above implication is T (think of the truth table for  $p \rightarrow q$ ). This case is not interesting!

Now suppose  $\exists x \forall y P(x, y)$  is T. Then there exists, say,  $x_0$  such that  $P(x_0, y)$  is always T for all  $y$ . Thus, for all  $y$ , there exists an  $x$ , namely  $x_0$ , such that  $P(x, y)$  is T. Therefore,  $\forall y \exists x P(x, y)$  is T.